ESOF 322

Software Engineering

LECTURE: 5

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OUTLINE:

- GitHub
  - SSH
  - Creating repo
  - Push, fetch, pull

GITHUB

WHAT IS GITHUB?

- A **cloud-based platform** for hosting Git repositories.
- Provides **collaboration tools**: pull requests, issues, code review.
- Integrates with **Continuous Integration (CI) / Continuous Delivery (CD)**, project management, and documentation.
- Owned by **Microsoft** since 2018.

GIT VS GITHUB

- **Git** = version control system (local).
- **GitHub** = remote hosting + collaboration features.
- GitHub uses Git under the hood.

GITHUB AUTHENTICATION METHODS

- **HTTPS**:
  - Login with **username + password** (deprecated for push).
  - Now requires **Personal Access Token (PAT)** instead of password.
- **SSH**:
  - Secure authentication using **SSH keys**.
  - Recommended for frequent use.
- **GitHub CLI** (gh)

AUTHENTICATION: UNIVERSITY VS. HOME PCS

On University Computers

- We'll use **HTTPS** with a **Personal Access Token (PAT)** after you sign in to the workstation with your university username and password.
- No SSH setup required on lab machines.

On Your Home Computer

- Run **two PowerShell commands** to start the **SSH agent**.
- **Add your SSH key** to the agent.
- Then you can **authenticate to your GitHub account over SSH**.

Note: The very first time you enable the SSH agent service on Windows, you may need to run PowerShell **as Administrator**.

GITHUB GENERATING SSH KEYS

https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent

- Command(Bash):

```bash
ssh-keygen -t ed25519 -C "your_email@example.com"
```

- Default location: `~/.ssh/id\_ed25519`.
- Add the public key to GitHub:
  - **Settings** → **SSH and GPG keys** → **New SSH key**.

ADDING AN SSH KEY TO GITHUB

- **Log in** to your GitHub account.
- Click your **profile icon** (top-right corner).
- Go to **Settings**.
- In the left sidebar, select **SSH and GPG keys**.
- Click **New SSH key**.
- Enter a **Title** (e.g., "My Laptop").
- Paste your **public key** (from ~/.ssh/id_ed25519.pub).
- Click **Add SSH key**.
- Confirm with your **GitHub password** or **2FA**.

ADDING YOUR SSH KEY TO THE SSH-AGENT

⚒ Adding your SSH key to the ssh-agent

`powershell`:

```
Get-Service -Name ssh-agent | Set-Service -StartupType Manual
Start-Service ssh-agent
ssh-add c:/Users/YOU/.ssh/id_ed25519
```

GITHUB SSH ADDING MULTIPLE DEVICES

- Each device should have its **own SSH key**.
- Add each key separately in GitHub settings.
- Helps track and revoke access per device.

HOW TO CREATE A NEW REPOSITORY ON GITHUB 1/2

1. **Log in** to your GitHub account.
2. Click the **"+" icon** in the top-right corner.
3. Select **"New repository"**.
4. Enter a **Repository name** (e.g., my-project).
5. (Optional) Add a **Description**.
6. Choose **Public** or **Private** visibility.

HOW TO CREATE A NEW REPOSITORY ON GITHUB 2/2

7. (Optional) Initialize with:
   - **README.md**
   - **.gitignore**
   - **License**
8. Click **Create repository**.
9. Copy the **SSH or HTTPS URL** to clone locally.

FROM MASTER TO MAIN – WHY THE CHANGE? 1/2

Background:

- Historically, Git’s default branch was named **master**.
- In recent years, the term was considered **politically and socially insensitive** in some contexts.

GitHub’s Response:

- GitHub changed the **default branch name** to **main** for new repositories.
- This is part of an **inclusive language initiative** across the tech industry.

FROM MASTER TO MAIN – WHY THE CHANGE? 2/2

What Does This Mean for You?

- When creating new repositories on GitHub, the default branch will be **main**.
- If your local repo still uses **master**, you can rename it:

```
git branch -m master main
```

THANK YOU!